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UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 11:
SUPERVISED LEARNING MODELLING AND EVALUATION: CLASSIFICATION USING DECISION TREES AND RANDOM FORESTS

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 - Sem 1

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Topic 9: Supervised Learning Modelling And Evaluation: Classification Using Decision Tree and Random Forest

- Introduction to Decision Tree
- Types of Decision Tree & Algorithms
- Terminology in Decision Tree
- Primary different between classification and regression tree
- Introduction to Random Forest
- Ensemble Learning in Random Forest
- Application of RF
- Advantages of DT & RF
- Disadvantages of DT & RF
- How to choose between DT & RF

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Introduction – Decision Tree

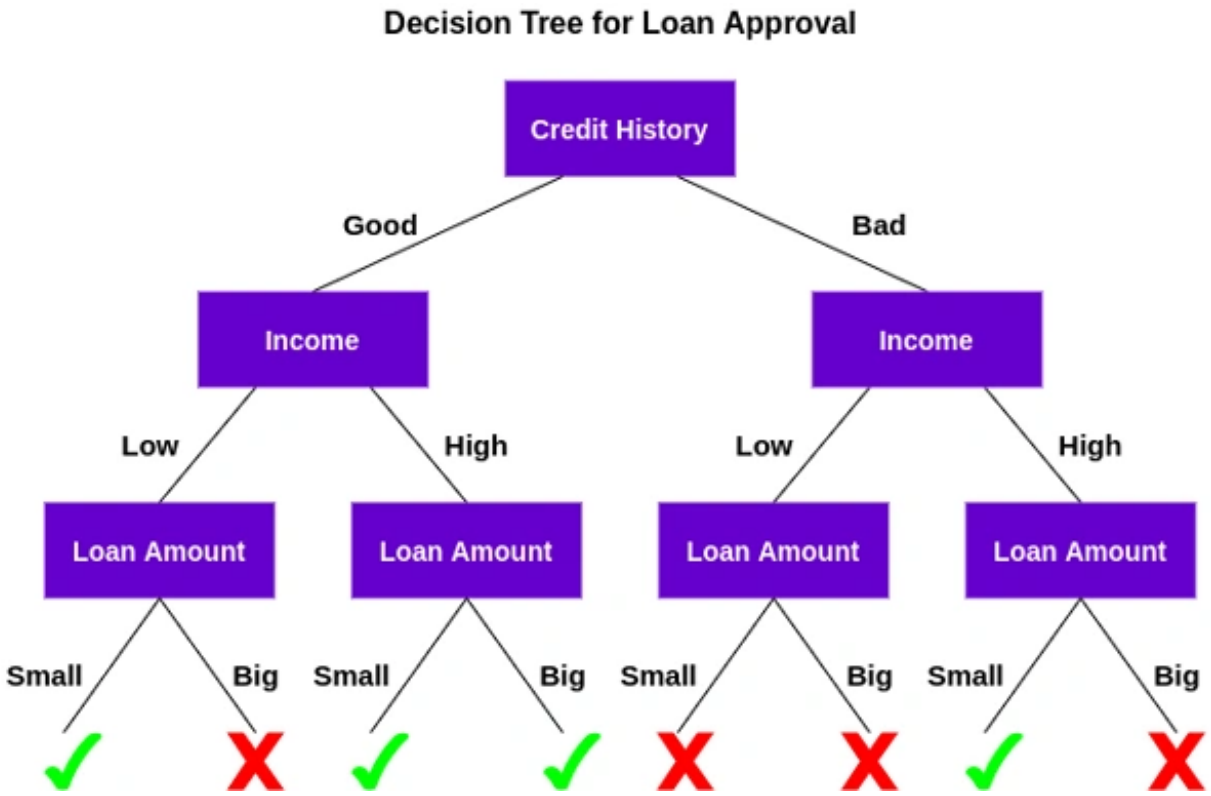
- Supervised machine learning algorithm.
- Classification and regression problems.
- Flow-chart like structure:

Nodes denotes the features, branches denoted rules, and leaves denotes the final result.

- Handle categorical and continuous data.

Introduction – Decision Tree

- Decision Tree for Loan Approval



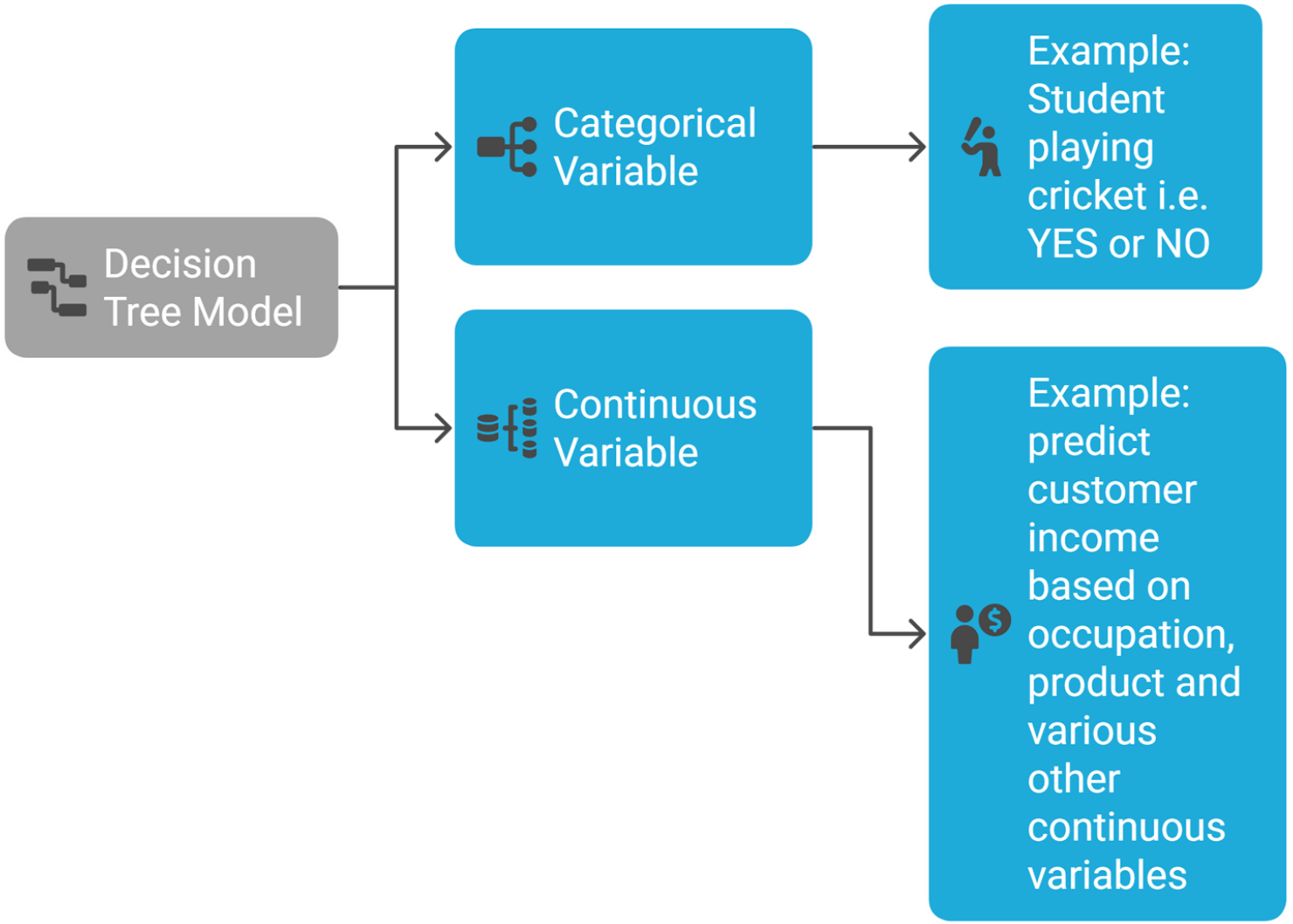
1. Checks customers' good credit history.
1. Classifies: customers with good credit history and customers with bad credit history.
1. Then, checks the customer's income.
1. Classifies the customer into: customers with good income and customers with bad income
1. Finally, it checks the loan amount requested by the customer.
1. Based on the outcomes from checking these three features, the decision tree decides whether the customer's loan should be approved.
1. Decision tree makes a series of decisions based on a set of features/attributes present in the data: credit history, income, and loan amount.

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Types - DT



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DT - Algorithms

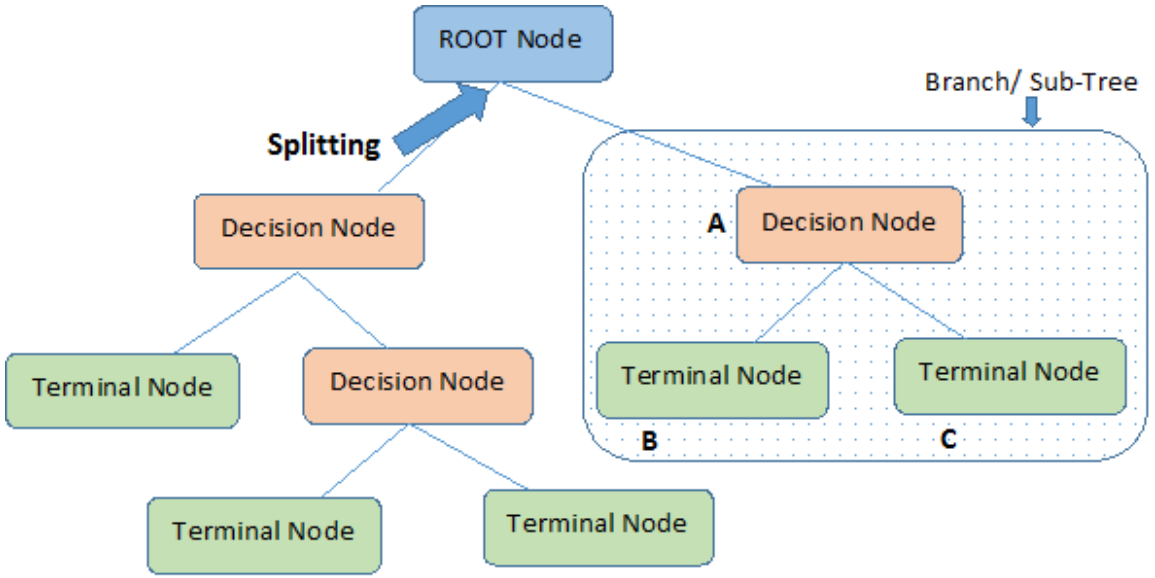
Feature/Aspect	ID3	C4.5	C5.0	CART
Developer	J. R. Quinlan	J. R. Quinlan	J. R. Quinlan	Breiman et al.
Tree Structure	Multiway	Multiway	Multiway	Binary
Target Type	Categorical only	Categorical	Categorical	Categorical & Numerical
Feature Type Support	Categorical only	Categorical & Continuous	Categorical & Continuous	Categorical & Continuous
Splitting Criterion	Information Gain	Gain Ratio (Information Gain normalized)	Boosted version of C4.5	Gini Index / MSE (for regression)
Pruning Method	Post-pruning	Post-pruning via rule simplification	Improved pruning and error handling	Cost-complexity pruning
Rule Extraction	Not supported	Converts to rule sets	Produces smaller and more accurate rule sets	Not supported
Performance	Baseline	Better accuracy and flexibility than ID3	Faster, less memory, more accurate than C4.5	Competitive performance, supports regression
Availability	Open-source	Open-source	Proprietary	Open-source

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Terminology - DT



Note:- A is parent node of B and C.

- **Root Node**: Represents entire population or sample and this further gets divided into two or more homogeneous sets.
- **Splitting**: Process of dividing a node into two or more sub-nodes.
- **Decision Node**: When a sub-node splits into further sub-nodes.
- **Leaf/ Terminal Node**: Nodes do not split.
- **Pruning**: Remove sub-nodes of a decision node, this process is called pruning. Opposite process of splitting.
- **Branch / Sub-Tree**: A sub section of entire tree.
- **Parent and Child Node**: A node, which is divided into sub-nodes is called parent node of sub-nodes where as sub-nodes are the child of parent node.

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CDT vs. RDT

Aspect	Classification Tree	Regression Tree
Target Variable Type	Categorical (e.g., Yes/No, Class A/B/C)	Continuous (e.g., Prices, Scores)
Prediction Value	Mode (most frequent class)	Mean (average value)
Used For	Classifying items into categories	Predicting numerical values
Splitting Criterion	Gini Index, Entropy, or Classification Error	Least Squares (minimizing variance in target values)
Output at Terminal Node	Class label (category)	Numerical value (average of responses)
Evaluation Metrics	Accuracy, Precision, Recall, F1-score	Mean Squared Error (MSE), R-squared
Example Use Case	Spam detection, Disease diagnosis	Predicting house prices, Sales forecasting

Topic 9: Supervised Learning Modelling And Evaluation: Classification Using Decision Tree and Random Forest

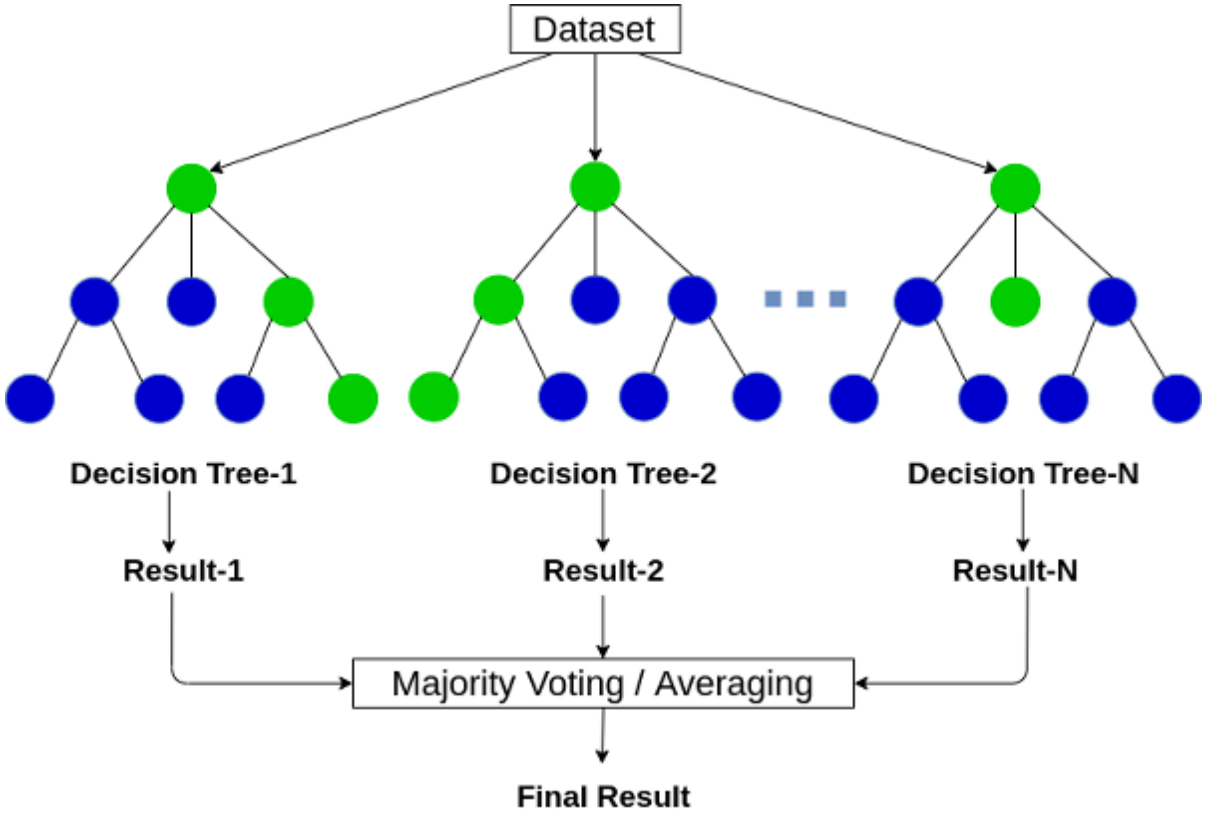
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Introduction – Random Forest

- A single tree is not sufficient for producing effective results. Thus, introduce Random Forest.
- Random Forest: tree-based algorithm that leverages the power of multiple decision trees to make decisions.
- Each node in the decision tree works on a random subset of features to calculate the output.
- The random forest then combines the output of individual decision trees to generate the final output.

Introduction – Random Forest



- Classification and regression task.
- Uses ensemble learning (combining multiple models/classifiers).
- As the number of decision tree in random forest increases the accuracy increases and overfitting also reduces.
- Bootstrapping is the process of randomly selecting items from the training dataset.
- This is a haphazard technique. It assembles randomized decisions based on several decisions and makes the final decision based on the majority voting.

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Introduction – Random Forest

- Three main hyperparameters: node size, the number of trees, and the number of features sampled.
- Made up of a collection of decision trees, and each tree in the ensemble is comprised of a data sample drawn from a training set with replacement, called the bootstrap sample.
- One-third of training set is set aside as test data, known as the out-of-bag (oob) sample.
- Another instance of randomness is then injected through feature bagging, adding more diversity to the dataset and reducing the correlation among decision trees.
- Depending on the type of problem, the determination of the prediction will vary.
- For a regression task, the individual decision trees will be averaged, and for a classification task, a majority vote.

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Ensemble Learning

- ‘Ensemble’ = group (of predictive models).
- A tree-based algorithm also suffers from the plague of bias and variance.
- Bias means, ‘how much on an average are the predicted values different from the actual value.’
- Variance means, ‘how different will the predictions of the model be at the same point if different samples are taken from the same population’.
- You build a small tree and you will get a model with low variance and high bias.
- How do you manage to balance the trade-off between bias and variance?

Types of Ensemble Learning

- Averaging: Taking the average of predictions from models in case of regression problem or while predicting probabilities for the classification problem.

Model1	Model2	Model3	AveragePrediction
45	40	65	50

- Majority vote: It's defined as taking the prediction with maximum vote / recommendation from multiple models' predictions while predicting the outcomes of a classification problem.

Model1	Model2	Model3	VotingPrediction
1	0	1	1

- Weighted average: In this, different weights are applied to predictions from multiple models then taking the average which means giving high or low importance to specific model output.

	Model1	Model2	Model3	WeightAveragePrediction
Weight	0.4	0.3	0.3	
Prediction	45	40	60	48

Ensemble Learning

Questions: If you create all the models on the same set of data and combine it, will it be useful? There is a high chance that these models will give the same result since they are getting the same input.

- Common techniques: Bagging, Boosting and Stacking.
- RF uses Bagging (or Bootstrap Aggregating) technique.
- Bootstrapping: sampling technique in which create subsets of observations from the original dataset, with replacement.
- The size of the subsets is the same as the size of the original set.
- The Bagging (or Bootstrap Aggregating) technique uses these subsets (bags) to provide a fair idea of the distribution (complete set).
- The size of subsets created for bagging may be less than the original set.

Ensemble Learning

Bagging: referred to as bootstrap aggregation.

To understand bagging, we first need to understand bootstrapping.

Bootstrapping is a sampling technique in which we choose ‘n’ observations or rows out of the original dataset of ‘n’ rows as well.

But the key is that each row is selected with replacement from the original dataset so that each row is equally likely to be selected in each iteration.

Let’s say we have 3 rows numbered 1, 2 and 3.

Data		Bootstraped Sample
Row 1		
Row 2		
Row 3		

Ensemble Learning

For bootstrapped sample, we choose one out of these three randomly. Say we chose Row 2.

Data	Bootstraped Sample
Row 1	Row 2
Row 2	
Row 3	

You see that even though Row 2 is chosen from the data to the bootstrap sample, it's still present in the data. Now, each of the three:

Rows have the same probability of being selected again. Let's say we choose Row 1 this time.

Data	Bootstraped Sample
Row 1	Row 2
Row 2	Row 1
Row 3	

Ensemble Learning

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Again, each row in the data has the same probability to be chosen for Bootstrapped sample. Let's say we randomly choose Row 1 again.

Data	Bootstraped Sample
Row 1	Row 2
Row 2	Row 1
Row 3	Row 1

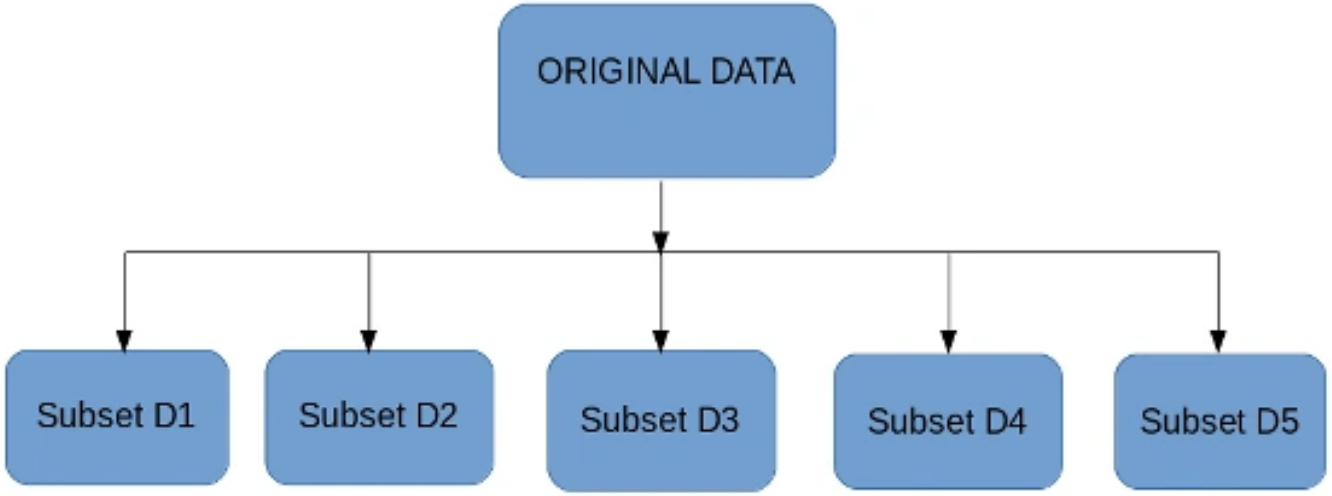
Thus, we can have multiple bootstrapped samples from the same data.

Once we have these multiple bootstrapped samples, we can grow trees and use the majority vote or averaging concepts to get the final prediction.

One important thing to note here is that it's done mainly to reduce the variance.

Now, random forest actually uses this concept but it goes a step ahead to further reduce the variance by randomly choosing a subset of features as well for each bootstrapped sample to make the splits while training.

Ensemble Learning



- Create multiple subsets from the original dataset by selecting observations with replacement.
- A base model (weak model) is created on each of these subsets.
- The models run in parallel and are independent of each other.

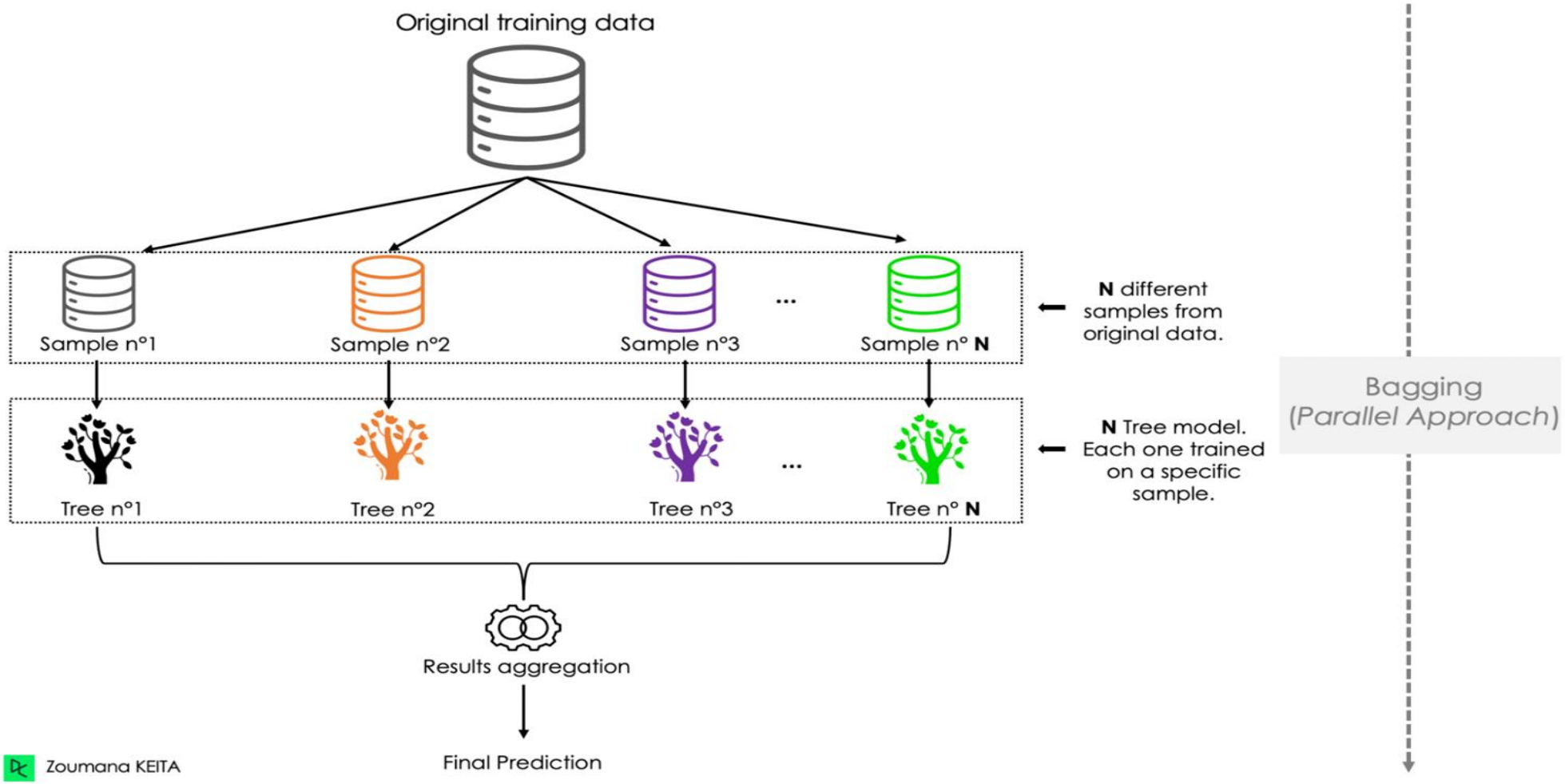
Ensemble Learning – Analogy 1

- Multiple traders - make a decision - the price of a stock will go up or not.
- All of them have a different understanding of the stock market - different mapping function from the problem statement to the desired outcome.
- Therefore, they are supposed to make varied predictions on the stock price based on their own understandings of the market.
- Now we can take all of these predictions into account while making the final decision.

Ensemble Learning – Analogy 2

- Candidate going through multiple rounds of job interviews.
- The final decision of candidate's ability is generally taken based on the feedback of all the interviewers.
- Although a single interviewer might not be able to test the candidate for each required skill and trait.
- But the combined feedback of multiple interviewers usually helps in better assessment of the candidate.

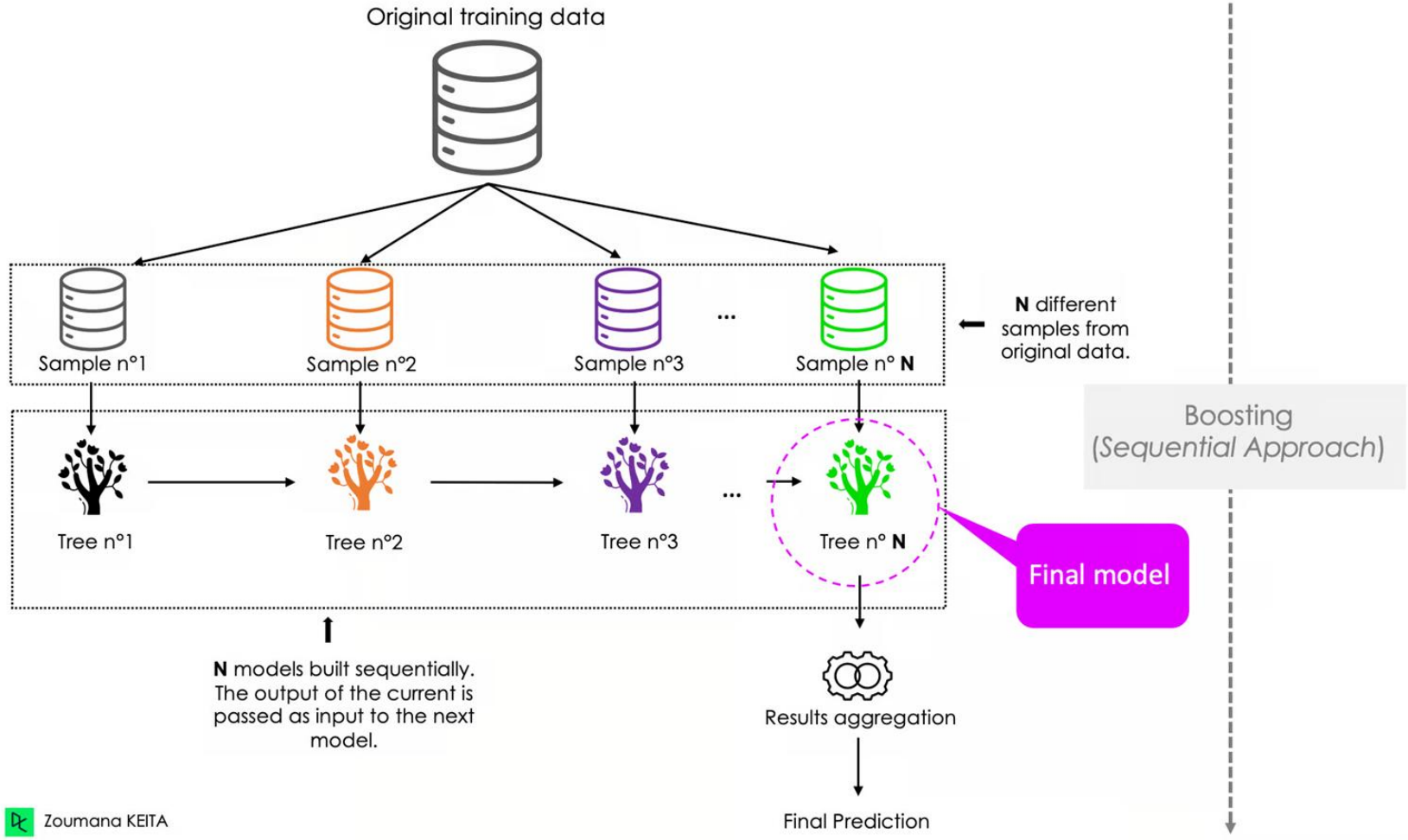
Bagging



Source: https://www.datacamp.com/tutorial/ensemble-learning-python

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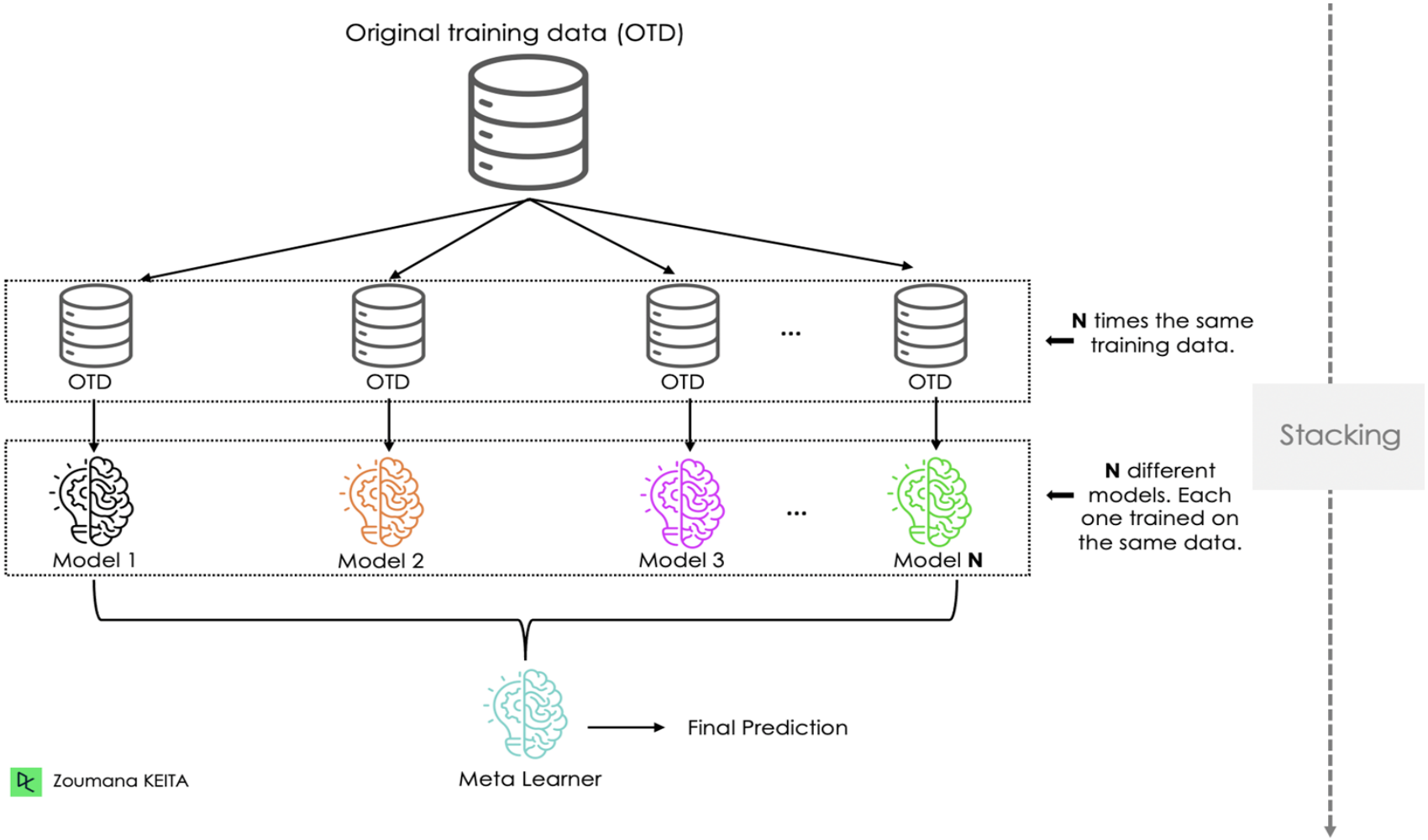
Boosting



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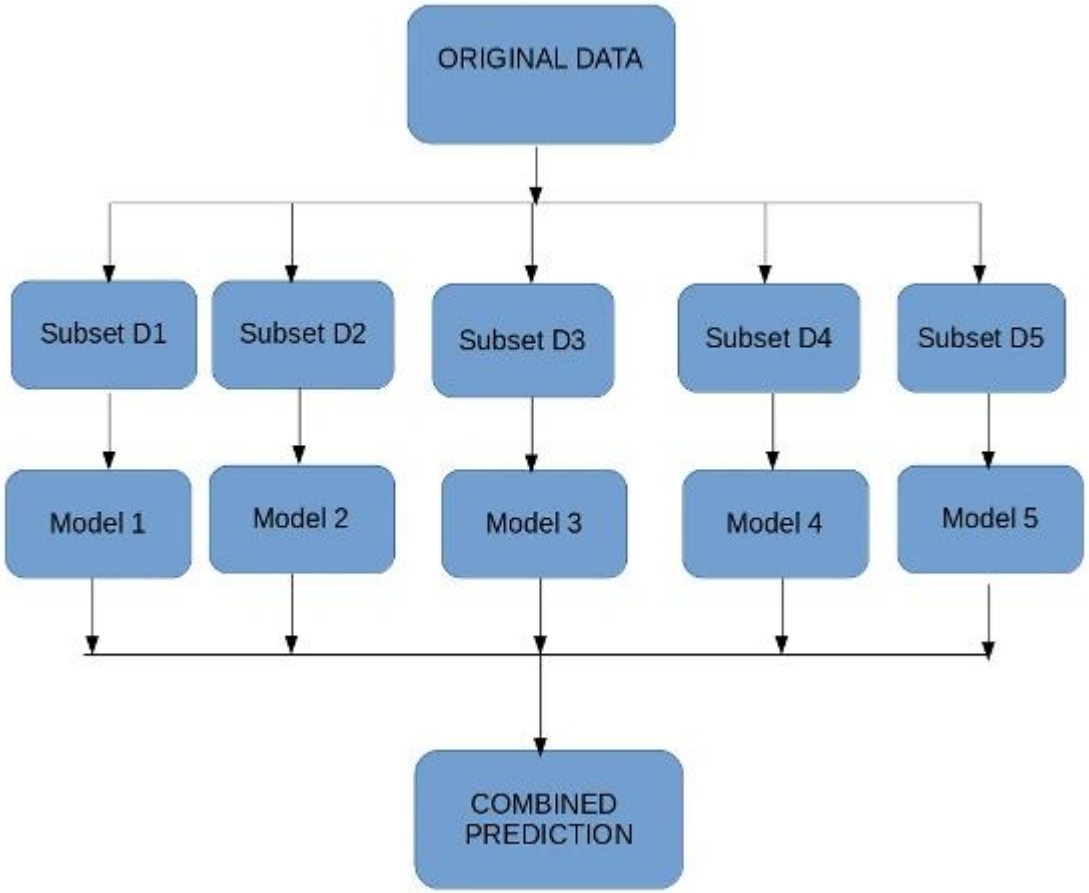
Stacking



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Random Forest – Ensemble Learning



- In general, ensembling is a technique of combining two or more algorithms of similar or dissimilar types called base learners / estimators.
- The base estimators in random forest are decision trees.
- The random forest randomly selects a set of features to decide the best split at each node of the decision tree.
- Looking at it step-by-step, this is what a random forest model does:

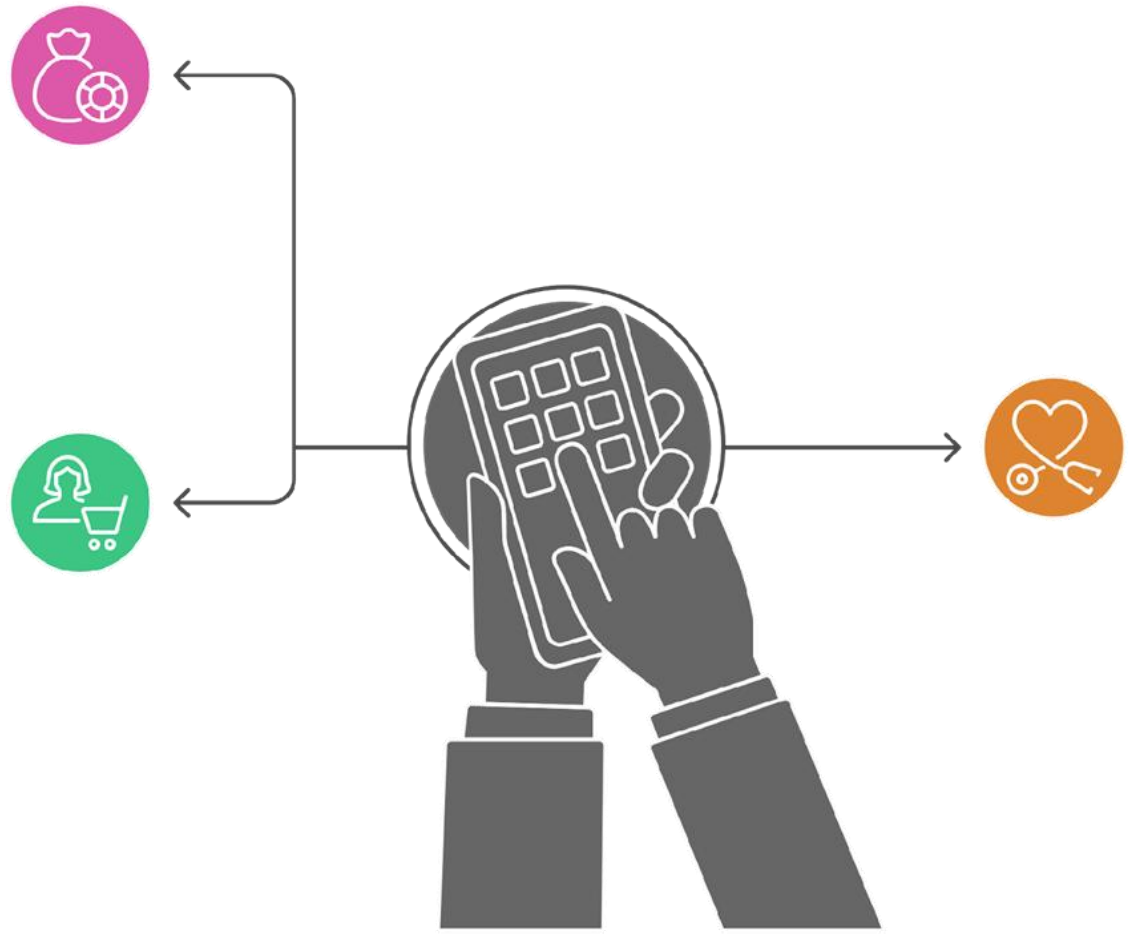
- Random subsets are created from the original dataset (bootstrapping).
- At each node in the decision tree, consider only a random set of features to decide the best split.
- A decision tree model is fitted on each of the subsets.
- Calculate the final prediction by averaging the predictions from all decision trees.

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Random Forest – Application



Finance

Evaluate credit risk and detect fraud. It also assists with option pricing.

E-commerce

Recommendation engines. Suggest products for cross-selling opportunities.

Healthcare

Computational biology for gene expression classification, biomarker discovery, and sequence annotation. Doctors can estimate drug response to specific medication.

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Advantages

Decision Tree	Random Forest
Simple to understand and to interpret. Trees can be visualised.	power of handle large data set with higher dimensionality.
Requires little data preparation.	effective method for estimating missing data and maintains accuracy when a large proportion of the data are missing.
Able to handle both numerical and categorical data.	balancing errors in data sets where classes are imbalanced.
Possible to validate a model using statistical tests.	Reduced risk of overfitting.
Performs well even if its assumptions are somewhat violated by the true model from which the data were generated.	Easy to determine feature importance. There are a few ways to evaluate feature importance. Gini importance and mean decrease in impurity (MDI).

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Disadvantages

Decision Tree	Random Forest
Overfitting.	Time-consuming process: Since can handle large data sets.
Decision trees can be unstable.	Requires more resources: process larger data sets.
Cannot guarantee to return the globally optimal decision tree.	More complex.
Decision tree learners create biased trees if some classes dominate.	Good at classification but not for regression problem: does not give precise continuous nature predictions. In case of regression, it doesn't predict beyond the range in the training data, and that they may over-fit data sets that are particularly noisy.
Not fit for continuous variables: While working with continuous numerical variables, decision tree-based classifiers loses information when it categorizes variables in different categories.	Black box approach for statistical modelers.

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How to choose?

Aspect	Decision Tree	Random Forest
Model Type	Single decision tree	Ensemble of multiple decision trees
Interpretability	Easy to interpret and understand	Harder to interpret due to multiple trees
Training Time	Faster to train	Slower to train, increases with number of trees
Prediction Accuracy	Lower accuracy, prone to overfitting	Higher accuracy, reduces overfitting
Overfitting	More prone to overfitting	Less prone to overfitting
Handling Large Datasets	Can handle but may not be optimal	Better suited for large datasets
Complexity	Simple structure	Complex structure
Use Case	Quick, interpretable insights, simple decisions	Predictive modeling requiring accuracy, less interpretability concerns
Feature Dependency	Can be unstable and dependent on a particular feature set	More stable due to aggregation of trees
Visualization	Simple, easy to visualize	Complex, challenging to visualize
Suitability for Beginners	More beginner-friendly	Requires more understanding of ensemble methods

How to choose?

- ✓ Use Decision Trees when you need fast, interpretable results and are working with smaller datasets or want to quickly understand data relationships.
- ✓ Use Random Forests when you want higher predictive accuracy, can afford longer training times, and are less concerned about interpretability.
- ✓ Use a decision tree when interpretability is important, and you need a simple and easy-to-understand model.
- ✓ Use a random forest when you want better generalization performance, robustness to overfitting, and improved accuracy, especially on complex datasets with high-dimensional feature spaces.
- ✓ If computational efficiency is a concern and you have a small dataset, a decision tree might be more appropriate.
- ✓ If you have a large dataset with complex relationships between features and labels, a random forest is likely to provide better results.

THANK YOU



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